

Part 4 of 4

Defence Lasers and Optronic Systems:

Gas Laser Electronics

Carbon dioxide and helium-neon lasers are the two commonly used gas lasers used in tactical military applications. High-power lasers, such as carbon dioxide gas dynamic lasers, hydrogen fluoride/deuterium fluoride lasers and chemical oxy-iodine lasers, which can generate mega-watts of CW power for directed-energy weapon applications, are also broadly classified as gas lasers. These lasers are pumped by gas dynamics or by chemical reactions. Focus in this concluding part of the article is on gas laser electronics with emphasis on the role of electronics in gas lasers having potential for tactical military applications

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Carbon dioxide laser-based range finders and laser radar and helium-neon laser-cavity-based ring-laser gyroscope inertial navigation sensors are among the common military applications exploiting use of gas lasers. In addition, such gas lasers as gas dynamic carbon dioxide lasers, chemical oxy-iodine lasers, all gas-phase iodine lasers, hydrogen fluoride and deuterium fluoride lasers, though not electrically or optically pumped, these are largely exploited to build high-power directed-energy weapons. We shall, however, confine our discussion in this article to electrically-excited gas lasers having military applications.

In both carbon dioxide and helium-neon lasers, the power supply used to initiate and subsequently sustain electrical discharge through the gas mixture contained in a sealed envelope constitutes the primary and essential component of electronics. The active medium is usually excited either by passing an electric discharge current along the length of the tube known as longitudinal excitation in both types of gas lasers (Fig. 1), or by an electric discharge perpendicular to the length of the laser tube known as transverse excitation which is common in carbon dioxide lasers only (Fig. 2). Frequency-stabilisation electronics used in the case of actively-stabilised Doppler

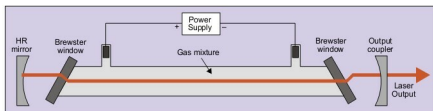


Fig. 1: Longitudinally-excited gas laser

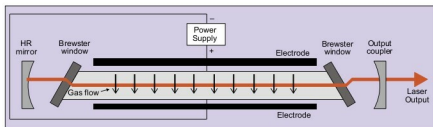


Fig. 2: Transversely-excited gas laser

broadened gas lasers, such as helium-neon and carbon dioxide, is another area that relates to gas laser electronics.

Negative resistance characteristics

Gas-discharge characteristics, when excited electrically, are the key to design of power supplies for gas lasers. Typical gas-discharge characteristics, as applicable to carbon dioxide and helium-neon lasers, exhibit negative resistance in their current-voltage relationship. The current is zero initially, which may rise to a few nano amperes after the applied voltage exceeds a certain threshold causing some ionisation of the gas mixture. This current increases slowly till it reaches the

breakdown point. At the breakdown point, large number of molecules in the gas mixture are ionised and conductivity increases significantly. Increase in discharge current causes further reduction in discharge resistance with the result that the voltage required to sustain the discharge actually decreases with increase in current. This gives rise to what is called negative resistance region in the gas-discharge current-voltage characteristics.

Helium-neon power supplies

The generalised form of a gas-laser power supply essentially comprises a high-voltage generation circuit that provides the starting voltage, either